SEMINAR REPORT

A seminar Report on

Phishing Website Detection Techniques

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960259	388563	Patil Vijaysing Vinod
2	24610960227	388585	Suryawanshi Nisha Kailas
3	24610960245	388552	Patil Lokesh Mangaldas
4	24610960252	388557	Patil Sagar Govind

FOR THE ACADEMIC YEAR : 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING S.S.V.P.S's
Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

DEPARTMET OF COMPUTER ENGINEERING

CERTIFICATE

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960259	388563	Patil Vijaysing Vinod
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of **CO5K** Semester of Diploma in **Computer Engineering** of **S.S.V.P.S's B.S.Deore Polytechnic, Dhule** (Inst.Code: 0059) has completed the term work satisfactorily in Course **Seminar Presentation and Project Initiation (Course Code: 315003)** for the academic year **2026 to 2027** as prescribed in the curriculum.

Place : Dhule

Date : .....

Signature:
Name : Shraddha S Hajari

Subject Teacher

Signature:
Name : Nishad D Patel

Head of the Department

Seal of Institution

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ABSTRACT

Unified Payments Interface (UPI) has become an important part of India's digital payment ecosystem. It enables users to transfer money between bank accounts using mobile applications, UPI IDs, QR codes, and other supported payment methods. The speed and convenience of digital payments also make fraud prevention and transaction monitoring important.

AI-Based Fraud Detection in UPI Payments is an approach that uses Artificial Intelligence (AI) and Machine Learning (ML) to examine transaction patterns and identify activities that may be suspicious. A detection model can consider features such as transaction amount, frequency, time, device information, account history, and unusual changes from a customer's normal behaviour, where such data is lawfully available.

Traditional fraud monitoring may depend on fixed rules and previously known patterns. Machine learning can learn from labelled examples and estimate the risk associated with a transaction. The output may support an alert, additional verification, or review by an authorized fraud-monitoring team. A model does not prove that a transaction is fraudulent, and its results must be combined with security controls and human oversight.

This seminar studies the UPI ecosystem, common fraud patterns, AI and ML techniques, system architecture, useful transaction features, evaluation methods, advantages, limitations, and future scope. It also proposes an educational prototype using a sample dataset rather than live banking data.

Keywords: UPI, Artificial Intelligence, Machine Learning, Fraud Detection, Digital Payments, Anomaly Detection, Cybersecurity.

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CHAPTER 1
INTRODUCTION

Digital payments have changed the way people pay for goods, services, bills, and person-to-person transfers. In India, the Unified Payments Interface (UPI) allows participating banks and payment applications to support fast account-to-account payments. Its convenience has encouraged widespread use by individuals, shops, businesses, and service providers.

At the same time, digital payment users may face fraud attempts such as social engineering, fake collect requests, impersonation, remote-access scams, malicious links, and misuse of compromised accounts. Many scams rely on persuading a user to approve a payment or disclose confidential information. Therefore, safe payment behaviour and effective monitoring are both important.

AI-based fraud detection uses data analysis and machine learning to identify transaction patterns that may indicate unusual or suspicious activity. Examples of signals include an unexpected transaction amount, many transfers in a short period, a new device, an unusual time, or a sudden change in the recipient pattern. These signals should be evaluated together rather than treated as proof on their own.

A well-designed detection system can assign a risk score and recommend an action such as allowing the transaction, requesting additional verification, or sending it for review. The system must protect personal data, avoid unnecessary disruption to genuine users, and be regularly evaluated against changing fraud patterns.

This seminar explains how AI can support UPI fraud detection and proposes a small educational model that can classify sample transactions as likely legitimate or potentially suspicious.

CHAPTER 2
LITERATURE REVIEW

The literature review examines existing approaches to digital payment fraud monitoring, machine learning-based anomaly detection, and responsible use of AI in financial services. It helps identify common methods, strengths, limitations, and research opportunities.

2.1 Digital Payment Fraud Monitoring

Payment systems use a combination of authentication, transaction limits, rule-based monitoring, risk controls, customer alerts, and investigations to reduce fraud. Rules can detect known patterns quickly, but static rules may not capture every new pattern. Monitoring must also distinguish malicious activity from unusual but genuine behaviour.

Main contribution: Establishes the need for layered security and continuous monitoring across the payment lifecycle.

Limitation: Fixed rules can generate false alerts or miss unfamiliar patterns if they are not updated.

2.2 Machine Learning for Financial Fraud Detection

Machine learning algorithms learn relationships from labelled examples or identify unusual behaviour without requiring every pattern to be written as a fixed rule. Supervised methods use transactions labelled as fraudulent or genuine. Unsupervised methods identify outliers or clusters that differ from typical behaviour.

Main contribution: Can analyze many transaction features and prioritize suspicious cases for further review.

Limitation: Model quality depends on reliable data, representative examples, and careful evaluation.

2.3 Anomaly Detection in Transaction Data

Anomaly detection attempts to find observations that differ from the normal pattern. In payments, anomalies may include unusually high amounts, rapid repeated transfers, changes in device characteristics, or new recipient relationships. Context is essential because a single unusual feature may be normal for a particular customer.

Main contribution: Can help identify previously unseen patterns when labelled fraud examples are limited.

Limitation: Many genuine transactions can look unusual, which may increase false positives.

2.4 Supervised Classification Techniques

Common classification algorithms include Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, and Support Vector Machines. These models learn from labelled transaction records and produce a class prediction or risk score. The choice of algorithm depends on dataset size, interpretability, speed, and evaluation results.

Main contribution: Enables measurable comparison using precision, recall, F1-score, and confusion matrices.

Limitation: Fraud examples are often much fewer than genuine transactions, creating class imbalance.

2.5 Deep Learning and Hybrid Approaches

Deep learning can model complex relationships in large datasets, while hybrid systems combine machine learning with rule-based checks and other security controls. A practical financial system may use multiple layers rather than relying on one model.

Main contribution: Flexible detection and the possibility of learning complex patterns.

Limitation: More complex models may require greater computing resources and can be difficult to explain.

2.6 Research Gap

AI-based payment fraud detection faces several open challenges. Fraud patterns change over time, while confirmed fraud labels may arrive after a delay. Genuine users may also make unusual payments, so models must limit false alarms while identifying a useful share of suspicious cases.

Another challenge is privacy. Transaction histories and device information can be sensitive, and a model should use only data that is necessary, authorized, and appropriately protected. Training data may be imbalanced, incomplete, outdated, or biased toward particular types of users and transactions.

Therefore, a useful educational study should compare a simple rule-based baseline with a machine learning model, evaluate more than accuracy, and clearly state that the model is a decision-support tool rather than a guarantee of fraud detection.

Research objectives from the literature

- Review common UPI-related fraud patterns and existing security controls.
- Study supervised learning and anomaly detection for transaction analysis.
- Identify transaction features that can be used safely in a sample dataset.
- Evaluate models using precision, recall, F1-score, and false-positive rate.
- Consider privacy, fairness, explainability, and changing fraud patterns.

TABLE NO. 2.1: LITERATURE REVIEW

Sr. No.	Research / Source	Approach	Key Objective / Contribution	Limitation / Note
1	RBI Annual Report 2024–25	Regulatory and sector overview	Discusses financial stability, cyber security, fraud monitoring, and responsible AI in finance.	Not a UPI-only model evaluation.
2	NPCI UPI Product Statistics	Payment ecosystem statistics	Provides official transaction-volume and value statistics for understanding UPI scale.	Statistics describe usage, not individual fraud risk.
3	NPCI, Conversational Payments on UPI whitepaper	Industry whitepaper	Discusses AI/ML patterns, payment verification, and fraud alerting in digital financial services.	Industry-level discussion; methods vary by provider.
4	RBI Annual Report 2023–24	Fraud analysis and regulation	Describes fraud trends and the importance of strengthening detection mechanisms.	Bank-level figures do not directly measure a single UPI model.
5	Bolton & Hand (2002), Statistical Fraud Detection: A Review	Research review	Explains statistical approaches for detecting financial fraud and the importance of data patterns.	Older review; payment technologies have evolved.
6	Phua et al. (2010), A Comprehensive Survey of Data Mining-based Fraud Detection Research	Survey	Reviews data-mining methods across several fraud domains.	Not limited to UPI transactions.
7	Chandola, Banerjee & Kumar (2009), Anomaly Detection: A Survey	Academic survey	Provides foundational methods for identifying anomalies in data.	Anomaly does not automatically mean fraud.
8	Saito & Rehmsmeier (2015), The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets	Evaluation research	Explains why precision-recall metrics are important for imbalanced classification.	Metric choice depends on the use case.
9	Breiman (2001), Random Forests	Ensemble learning research	Introduces a tree-based ensemble method that can model nonlinear feature relationships.	Requires suitable training and validation.
10	Lundberg & Lee (2017), A Unified Approach to Interpreting Model	Explainable AI research	Introduces SHAP values to explain model predictions.	Explanations do not prove causality.

	Predictions			
11	Python scikit-learn documentation	Technical documentation	Describes classifiers, data preprocessing, metrics, and model validation tools.	Implementation guidance, not evidence of real-world UPI performance.
12	RBI, Annual Report 2024–25, Chapter VI	Official annual report	Highlights responsible AI and strengthened fraud detection as financial-sector priorities.	Broad sector context; does not specify a universal fraud algorithm.

Table 2.1 summarizes selected official and industry sources relevant to the seminar. These sources provide context for digital payments, fraud prevention, and the role of AI; they do not imply that every UPI application uses the same detection model.

CHAPTER 3
INFORMATION

3.1 Fundamentals of UPI and Payment Fraud

Unified Payments Interface is an instant payment system that supports bank-to-bank transfers through participating institutions and payment applications. Users may initiate payments using a UPI ID, mobile number linked to a supported account, QR code, or other available flow. Authentication and authorization are handled through the participating payment ecosystem.

Payment fraud can involve deception, account compromise, unauthorized access, or manipulation of a user into approving a transfer. Some common patterns are:

1. Social engineering: a fraudster persuades a user to approve a payment or reveal confidential information.
2. Fake collect request: a user is misled into approving a request to send money.
3. Impersonation: a fraudster pretends to be a trusted person, business, or official.
4. Remote-access scam: a user is tricked into installing or sharing access through a remote-support tool.
5. Mule-account activity: accounts may be used to receive and move proceeds of fraud.

A detection system can identify suspicious transaction patterns, but it cannot prevent every scam. User education, secure authentication, transaction controls, and prompt reporting remain essential.

3.2 AI and Machine Learning Techniques

1. Rule-Based Detection

Rule-based detection checks predefined conditions, such as repeated transactions within a short interval or an amount that exceeds a configured threshold. Rules are simple and explainable, but they must be maintained and may miss new patterns.

2. Supervised Machine Learning

Supervised learning uses historical records labelled as genuine or fraudulent. The model learns associations between input features and labels, then predicts a class or score for unseen records. Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting are common candidates.

3. Unsupervised Anomaly Detection

Unsupervised methods identify records that differ from the majority without requiring every record to have a fraud label. Isolation Forest and clustering-based methods are examples. Unusual activity must be reviewed carefully because anomalies may be legitimate.

4. Hybrid Detection

A hybrid design combines business rules, machine learning, and human review. For example, a model may calculate a risk score while rules enforce transaction limits and authentication requirements.

3.3 Model Selection

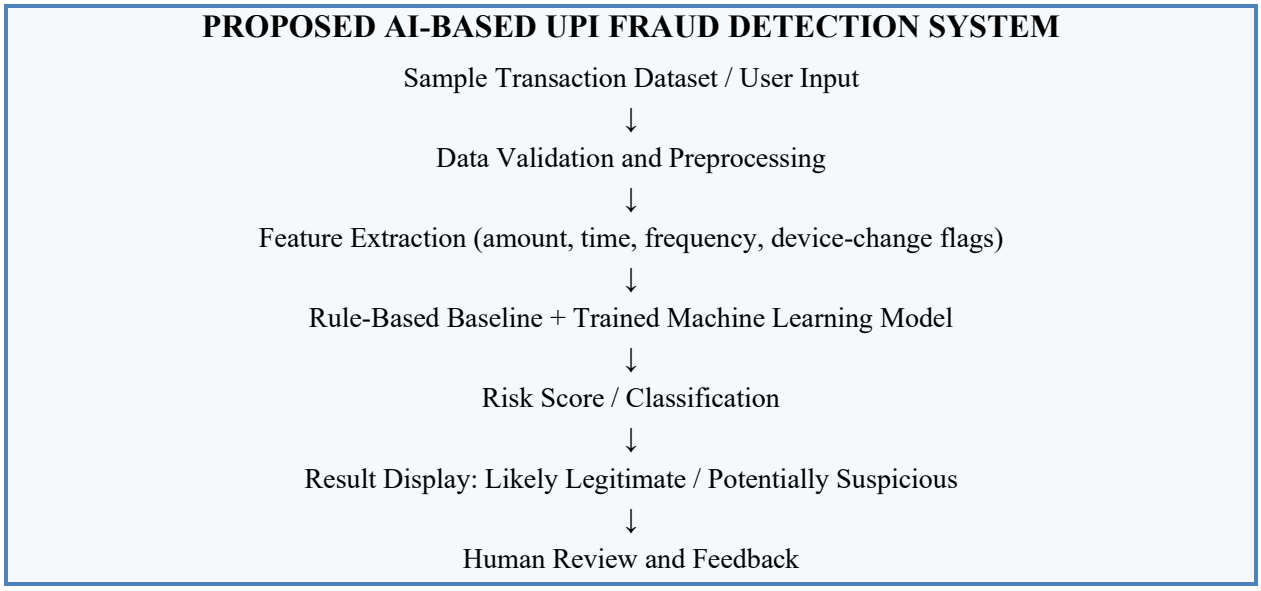
For a diploma-level prototype, Random Forest or Logistic Regression can be compared with a simple rule-based baseline. The final choice should be based on validation results, clarity, training time, and the cost of false alerts and missed fraud. Accuracy alone is not enough when fraud cases are rare.

TABLE NO. 3.1: TECHNOLOGY PURPOSE

Sr. No.	Technology	Purpose
1	Python	To implement data processing and machine learning.
2	Visual Studio Code	To write, edit, and run Python programs.
3	Pandas	To load, organize, and process transaction datasets.
4	NumPy	To perform numerical operations.
5	Scikit-learn	To train, test, and evaluate machine learning models.
6	Matplotlib / Seaborn	To visualize distributions and model evaluation results.
7	Streamlit	To create a simple demonstration interface.
8	CSV Dataset	To store sample transaction features and labels.
9	Jupyter Notebook	To explore the dataset and test models.
10	Microsoft Word / WPS Office	To prepare and format the seminar report.
11	Microsoft PowerPoint / WPS Presentation	To prepare seminar slides.
12	Web Browser	To access official documentation and research sources.

3.4 SYSTEM ARCHITECTURE

The proposed educational system receives sample transaction details, validates the input, prepares selected features, and passes them to a trained model. The model returns a risk classification that is shown to the user. This design is for demonstration only and is not connected to any live payment service.



Working explanation

6. A sample transaction record is selected from a dataset or entered into the demo interface.
7. The system checks that required fields are present and converts data into the model's expected format.
8. The model evaluates the available features and produces a class prediction or risk score.
9. The interface displays the result with a clear explanation that it is an estimate, not proof.
10. For a real financial service, any response would require authorized infrastructure, security review, privacy controls, and institution-approved procedures.

TABLE NO. 3.2: TRANSACTION FEATURES USED FOR DETECTION

Feature	Description	Example of a possible signal
Transaction amount	Value of the payment.	Amount far outside a user's usual range.
Transaction frequency	Number of transactions in a time window.	Many transfers in a short interval.
Time of transaction	Hour or time category.	Activity at an unusual time compared with history.
Recipient novelty	Whether the recipient has been seen before.	First transfer to a new recipient.
Device change flag	Whether the device differs from known activity, if available lawfully.	New device plus other unusual indicators.
Account activity history	Summary of past transaction patterns.	Sudden change in amount or frequency.
Failed attempt count	Recent failed or repeated attempts, if recorded.	Repeated attempts that may need review.
Geographic consistency	Coarse location consistency where available and permitted.	Unexpected change, interpreted cautiously.

No single feature can reliably determine fraud. A large payment or a new recipient may be completely genuine. A model should combine signals, account for context, protect sensitive information, and be evaluated for false positives and false negatives.

3.5 Evaluation of Detection Systems

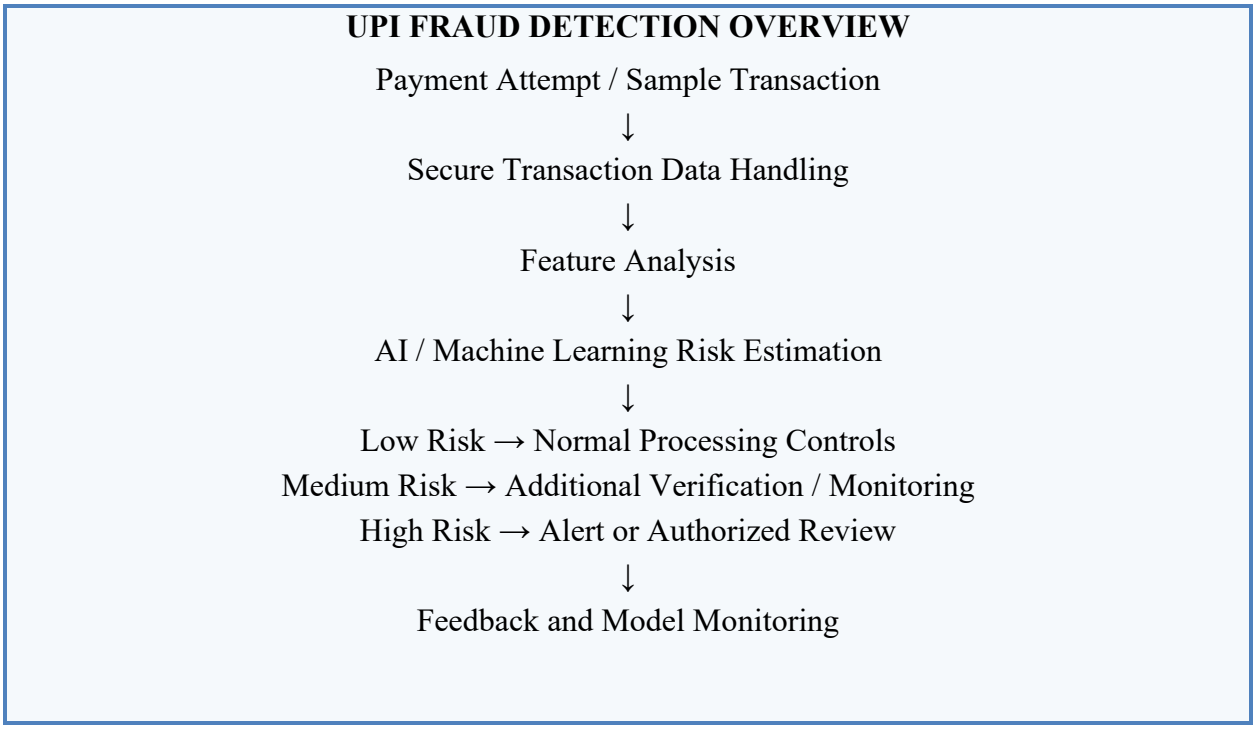
A fraud detection model should be evaluated on a held-out test set that was not used to train the model. Where possible, a time-based split should be considered so that evaluation resembles the challenge of predicting future transactions.

- Accuracy: the proportion of all predictions that are correct.
- Precision: among transactions predicted as suspicious, the proportion that are actually labelled suspicious.
- Recall: among all labelled fraudulent transactions, the proportion detected by the model.
- F1-score: a combined measure of precision and recall.
- False-positive rate: the proportion of genuine transactions incorrectly flagged.
- Confusion matrix: summarizes true positives, true negatives, false positives, and false negatives.

Actual / Predicted	Predicted Fraud	Predicted Genuine
Actual Fraud	True Positive (TP)	False Negative (FN)
Actual Genuine	False Positive (FP)	True Negative (TN)

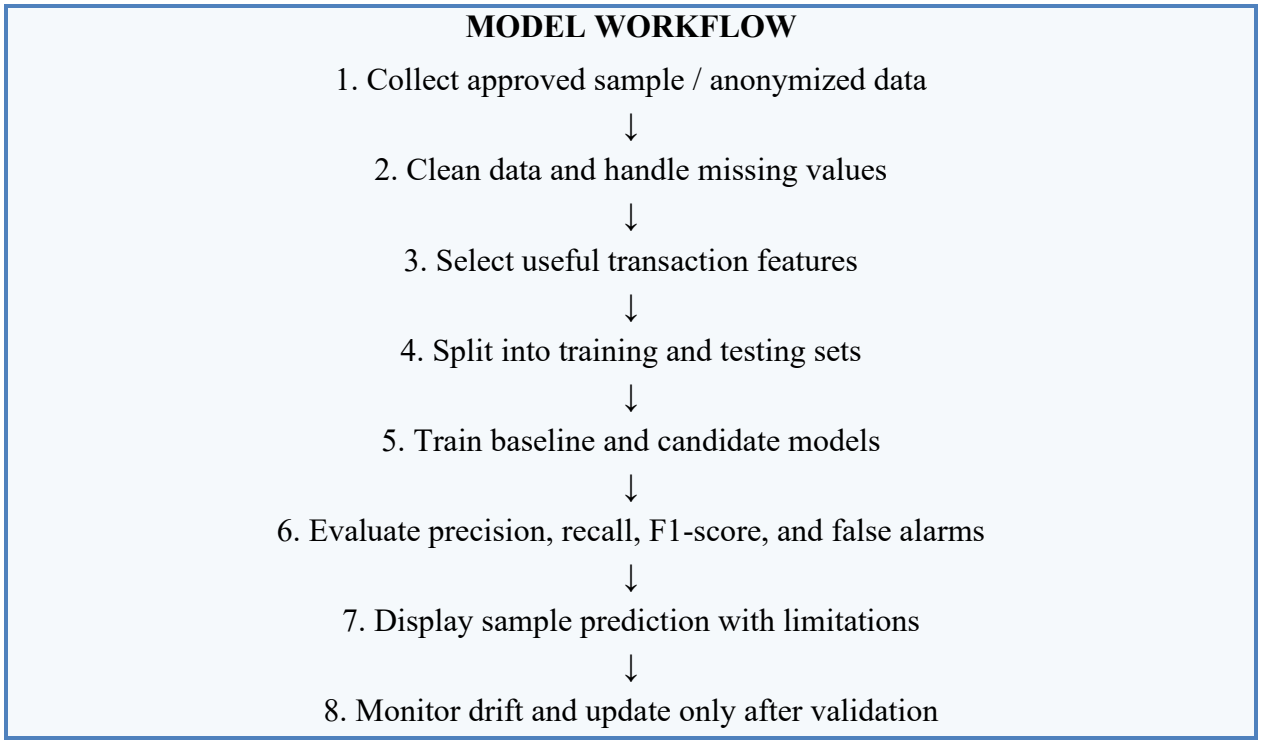
Accuracy can be misleading when fraudulent transactions are rare. A useful evaluation reports multiple metrics and considers the operational cost of both missed fraud and unnecessary alerts. Any performance figure in a student report should be measured from the actual test data, not invented.

FIG. NO. 3.1: OVERVIEW OF AI-BASED UPI FRAUD DETECTION



The exact action for a real payment must be decided by the authorized payment provider and its risk policy. A student prototype should only display a simulated result and must not attempt to block, initiate, or alter real payments.

FIG. NO. 3.2: WORKING FLOW OF A FRAUD DETECTION SYSTEM



Important safeguards

- Never collect or store real UPI PINs, OTPs, passwords, or bank login credentials.
- Use synthetic or appropriately anonymized data for the academic demonstration.
- Do not connect the student prototype to live banking APIs or real payment accounts.
- Keep predictions explainable and make clear that a warning is not a final fraud determination.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 Advantages

1. Early risk identification: may highlight unusual transaction patterns for review.
2. Automated analysis: can process many records consistently.
3. Adaptability: models can be retrained when validated new patterns become available.
4. Risk prioritization: helps investigators focus on higher-risk cases.
5. Pattern recognition: can combine multiple weak signals rather than rely on one fixed rule.
6. Decision support: can complement authentication, user education, and established security controls.
7. Research value: allows comparison of algorithms and evaluation metrics using a sample dataset.

4.2 Disadvantages

1. False positives: genuine payments may be flagged as suspicious.
2. False negatives: some fraudulent transactions may not be detected.
3. Data dependency: performance depends on representative, reliable, and current data.
4. Class imbalance: fraud cases may be much fewer than genuine transactions.
5. Privacy concerns: transaction data can reveal sensitive personal behaviour.
6. Changing tactics: fraud patterns evolve and models may become less effective.
7. Explainability challenges: complex models may be harder to interpret.

CHAPTER 5
CONCLUSION

UPI has made digital payments fast and convenient, but fraud prevention remains an important part of a trusted payment ecosystem. Fraud may involve social engineering, impersonation, account compromise, or unusual transaction patterns. Security therefore requires a combination of safe user behaviour, authentication, transaction controls, monitoring, and effective reporting.

Artificial Intelligence and Machine Learning can support fraud monitoring by examining patterns across transaction features and estimating risk. Supervised classifiers can learn from labelled examples, while anomaly detection can identify records that differ from expected behaviour. Hybrid approaches can combine model scores with predefined rules and human review.

However, AI is not a complete solution. Models can make mistakes, data may be imbalanced or outdated, and false alarms can inconvenience genuine users. Privacy, fairness, explainability, and ongoing evaluation are essential. Model performance should be measured using precision, recall, F1-score, false-positive rate, and a confusion matrix rather than accuracy alone.

In conclusion, AI-based fraud detection is a promising decision-support approach for digital payments. A student prototype can demonstrate the basic workflow using sample data, but real-world deployment requires authorized data access, secure infrastructure, regulatory compliance, monitoring, and review by qualified financial-sector professionals.

CHAPTER 6
PROJECT INITIATION

6.1 Description of Problem Statement

UPI payments are used for person-to-person transfers, merchant payments, bill payments, and other digital transactions. Fraudsters may exploit user trust, compromised accounts, or unusual transaction activity to attempt unauthorized transfers. A monitoring system must identify suspicious patterns quickly while avoiding excessive disruption to genuine customers.

Traditional rule-based systems are useful for known risks but may not identify every new or changing pattern. Machine learning can analyze multiple transaction features and provide a risk estimate; however, it depends on data quality and may produce false positives or false negatives.

Problem statement: To study how Artificial Intelligence and Machine Learning can be used to analyze sample UPI transaction features, identify potentially suspicious patterns, and support fraud-risk alerts while considering accuracy, privacy, and limitations.

6.2 Scope and Objectives

Scope:

- Study UPI payment flow and common categories of payment fraud.
- Review rule-based, supervised, and anomaly-detection approaches.
- Identify safe and useful features in a sample transaction dataset.
- Build or describe a small educational classification prototype.
- Evaluate the model using appropriate classification metrics.
- Discuss privacy, false alerts, explainability, and future improvements.

Objectives:

11. Understand AI-based fraud detection in digital payments.
12. Study selected machine learning algorithms for transaction classification.
13. Prepare sample data and extract relevant features.
14. Compare a simple baseline with a machine learning model.
15. Measure precision, recall, F1-score, and false-positive rate.
16. Explain the limitations and safeguards required for responsible use.

6.3 Stakeholders

Sr. No.	Stakeholder	Role and Responsibility
1	UPI Users	Use digital payment services and follow safe payment practices.
2	Banks and Payment Service Providers	Operate payment services, apply authorized risk controls, and investigate reported fraud.
3	NPCI and Payment Ecosystem Participants	Maintain ecosystem rules, standards, and operational processes within their responsibilities.
4	Fraud Monitoring Teams	Review alerts, investigate suspicious cases, and provide feedback.
5	Cybersecurity Professionals	Assess security risks and protect systems and data.
6	Data Scientists / ML Engineers	Prepare data, train models, evaluate results, and monitor model quality.
7	Regulators	Set applicable requirements and supervise regulated entities.
8	Students and Researchers	Study methods using safe, authorized datasets and document limitations.
9	Educational Institution	Provide guidance and assess the seminar and project work.
10	Software Developers	Build secure interfaces and maintain system functionality.

6.4 Platform / Equipment / Resources Identification

TABLE NO. 6.1: PLATFORM

Sr. No.	Platform	Purpose
1	Windows Operating System	To run development software and prepare the seminar project.
2	Python	To implement data processing and ML algorithms.
3	Visual Studio Code / Jupyter Notebook	To write and test code.
4	Microsoft Word / WPS Office	To prepare and format the report.
5	Microsoft PowerPoint / WPS Presentation	To prepare seminar slides.
6	Google Scholar	To search for academic research papers.
7	Web Browser	To access official documentation and reference material.

TABLE NO. 6.2: EQUIPMENT

Sr. No.	Equipment	Purpose
1	Laptop / Computer	To research, code, and prepare the report.
2	Internet Connection	To access documentation, papers, and sample datasets.
3	Smartphone	To review payment safety examples and the presentation.
4	Projector	To display seminar slides.
5	Pen Drive / External Storage	To store and transfer project files.
6	Power Supply / Charger	To keep the laptop operational during project work.

TABLE NO. 6.3: RESOURCES

Sr. No.	Resource	Purpose
1	Research Papers	To study existing fraud-detection and anomaly-detection methods.
2	Sample Transaction Dataset	To train and test a student-level model.
3	Python Libraries	To process data and implement detection algorithms.
4	Pandas and NumPy	To organize, transform, and analyze data.
5	Scikit-learn	To train and evaluate machine learning models.
6	Matplotlib	To visualize class distribution and evaluation results.
7	Technical Documentation	To understand algorithms, libraries, and metrics.
8	Official UPI / RBI Resources	To understand the payment ecosystem and security context.
9	Computer and Internet	To conduct research and prepare project material.

The prototype should use synthetic or public anonymized transaction data. Real UPI PINs, OTPs, passwords, bank credentials, and identifiable transaction histories must not be collected for this academic project.

CHAPTER 7

SEMINAR PRESENTATION

S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering
Academic Year : 2026-2027

SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	SPI – 315003	Date :

Topic : Using AI for Fraud Detection in UPI Payments

Roll No.	Name of Student
76	Vijaysing Vinod Patil
98	Nisha Kailas Surywanshi
65	Lokesh Mangaldas Patil
70	Sagar Govind Patil

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Abstract

- UPI enables fast digital payments, but it can also be misused for fraud. This project explores how Artificial Intelligence (AI) and Machine Learning (ML) can identify suspicious UPI transactions by analysing transaction amount, frequency, device details, location patterns and transaction history.
- A detection model can assign a risk score and flag unusual activity for further verification. The aim is to support faster fraud detection, reduce financial losses and make digital payments safer.
- Final decisions should include secure verification and human review where required.

3

1. Introduction

- UPI is a widely used system for instant digital payments in India.
- Fraud may include phishing, social engineering, fake payment requests and account misuse.
- Traditional rule-based checks may miss new or changing fraud patterns.
- AI can learn from transaction data and identify unusual behaviour.
- The system can flag risky transactions for additional verification.
- The goal is to improve fraud detection while keeping genuine payments convenient.

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2. Literature Survey

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Rule-Based Fraud Detection	Rules, SQL, transaction monitoring	Amount, time, transaction count	Flag transactions matching known rules	May miss new fraud patterns
2	Machine Learning Classification	Decision Tree, Random Forest	Labelled transaction records	Classify transactions as genuine or suspicious	Needs quality labelled data
3	Anomaly Detection	Isolation Forest, clustering	Transaction behaviour features	Find unusual transaction patterns	Can flag genuine unusual activity
4	Neural Network Detection	Deep Learning, neural networks	Historical transaction features	Learn complex fraud patterns	Needs more data and computing
5	Behaviour-Based Detection	ML, user behaviour analytics	User's normal payment patterns	Detect sudden behaviour changes	Behaviour may change legitimately
6	Real-Time Transaction Scoring	Streaming data, ML model	Live transaction features	Score risk before completing payment	Low latency is essential

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	Graph-Based Fraud Analysis	Graph analytics, ML	Accounts, devices, payment links	Identify connected fraud networks	Graph data can be complex
8	Hybrid Fraud Detection	Rules + ML models	Transaction and rule alerts	Combine known rules with learned patterns	Requires careful integration
9	SMS / Phishing Risk Analysis	NLP, text classification	Reported messages and URLs	Identify suspicious payment lures	May miss new scam wording

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Device and Session Signals	Device fingerprinting, analytics	Device, session and login signals	Spot unusual device or access activity	Privacy and spoofing concerns
11	Imbalanced Data Handling	Sampling, class weights, metrics	Fraud-heavy imbalanced datasets	Improve learning from rare fraud cases	May increase false alarms
12	Explainable Fraud Detection	Feature importance, explainable AI	Model outputs and transaction features	Explain why a transaction was flagged	Explanations may be imperfect

3. Information to Seminar Topic

- Technology: Artificial Intelligence and Machine Learning
- Main Purpose: Detect potentially fraudulent UPI payments.
- Key Concept: Analyse transaction patterns and user behaviour.
- Approach: Extract features, score transaction risk and flag anomalies.
- Main Benefit: Earlier warnings and fewer missed suspicious transactions.
- Application: Payment platforms, banks and fraud monitoring teams.

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4. Advantages / Disadvantages

- **Advantages**
- Detects unusual transaction patterns quickly.
- Can adapt to emerging fraud trends when retrained.
- Supports real-time risk scoring and alerts.
- Can reduce manual review workload.
- **Disadvantages**
- Needs reliable and representative data.
- False alerts may inconvenience genuine users.
- Fraud patterns and attacker methods keep changing.

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5. Conclusion

- AI-based fraud detection can help make UPI payments safer by learning patterns from transaction data and identifying unusual activity.
- Machine Learning models can support risk scoring and alert payment providers before or after a suspicious payment. The system's effectiveness depends on good data, regular model updates, secure handling of information and evaluation of false positives and false negatives.
- AI should support existing security controls rather than replace user verification and responsible review.

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6. Project Initiation

- The project is initiated to address the risk of fraudulent UPI transactions and the limitations of detecting every new scam using fixed rules alone.
- It proposes an AI/ML approach that analyses transaction features, learns typical patterns and flags activity that appears suspicious.
- The project will study suitable algorithms, data requirements, evaluation measures and the need for privacy and security. The expected outcome is a conceptual fraud-detection workflow that can help payment teams prioritise risky cases.

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6.1 Description of Problem Statement

- UPI payments are fast and convenient, but users may face phishing, social engineering, fake collect requests and unauthorised transactions.
- Fixed rules may not detect every new fraud pattern and manual investigation can take time. The problem is to identify suspicious transactions accurately and quickly while avoiding unnecessary interruption of genuine payments.
- An AI-based approach can analyse transaction patterns and generate risk alerts for further verification.

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6.2 Scope and Objectives

- **Objectives**
- - Study common UPI fraud patterns.
- - Identify useful transaction features.
- - Explore ML methods for fraud classification.
- - Flag unusual payments using a risk score.
- - Evaluate precision, recall and false alarms.

- **Scope**
- - Focus on analysis of UPI transaction patterns.
- - Use authorised or anonymised sample data.
- - Compare rule-based and AI-based approaches.
- - Provide alerts for review, not automatic accusations.
- - Consider privacy, security and model updates.

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6.3 Stakeholders

- UPI Users: Need safe payments and clear alerts.
- Banks and Payment Apps: Monitor and investigate suspicious transactions.
- NPCI / Payment Ecosystem: Maintain secure and reliable UPI services.
- Fraud Analysts: Review flagged activity and investigate cases.
- AI / Data Teams: Develop, test and monitor detection models.
- Security and Compliance Teams: Protect data and ensure responsible use.

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6.4 Seminar Execution Plan

- 1. Current Scenario: UPI supports instant digital payments.
- 2. Problem: Scams and unauthorised payments can cause losses.
- 3. Existing Approach: Rules, limits, alerts and manual review.
- 4. Research Gap: New fraud patterns may bypass fixed rules.
- 5. Proposed Study: Apply ML to transaction features and risk scoring.
- 6. Evaluation: Check precision, recall and false-positive rate.
- 7. Future Work: Test with approved data and improve real-time alerts.

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6.5 Platform / Equipment / Resources

- Hardware Requirements:
- - Laptop or desktop computer.
- - 4 GB RAM minimum; 8 GB recommended.
- - Internet connection.

- Software / Tools:
- - Python and Jupyter Notebook.
- - Pandas and NumPy for data handling.
- - Scikit-learn for ML algorithms.
- - Matplotlib / Seaborn for analysis.

- Other Resources:
- - Public or anonymised transaction dataset.
- - Model evaluation and testing tools.
- - Secure data storage and documentation.

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- **Online Documentation**
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- Research / Technical Resources

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5. Reserve Bank of India — payment system and safety publications.
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7. Use only authorised, anonymised datasets for testing.

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Thank You

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6. NumPy Developers. NumPy Documentation. Available at: https://numpy.org/doc/

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9. Reserve Bank of India. Annual Report 2024–25. Official RBI website.

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960259	388563	Patil Vijaysing Vinod
2	24610960227	388585	Suryawanshi Nisha Kailas
3	24610960245	388552	Patil Lokesh Mangaldas
4	24610960252	388557	Patil Sagar Govind

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388493	388512	388519	388524
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : ----------

(Course Expert/s)